Project Final Report

Date	18 Feb 2026
Team ID	LTVIP2026TMIDS73343
Project Name	Heart Disease Analysis
Maximum Marks	5 Marks

1. INTRODUCTION

Project Overview

This project focuses on analyzing heart disease data to understand the main factors that increase health risks.

- I used a large clinical dataset with 4,500 patient records to find the key reasons behind heart disease.
- The technology used in this project includes MySQL for storing and managing data, Tableau for creating visualizations, and Flask for building the web application.
- The final result of the project includes 10 interactive charts and 2 detailed dashboards that allow users to explore health risks easily.

Purpose of the Project

This section explains why I created this project.

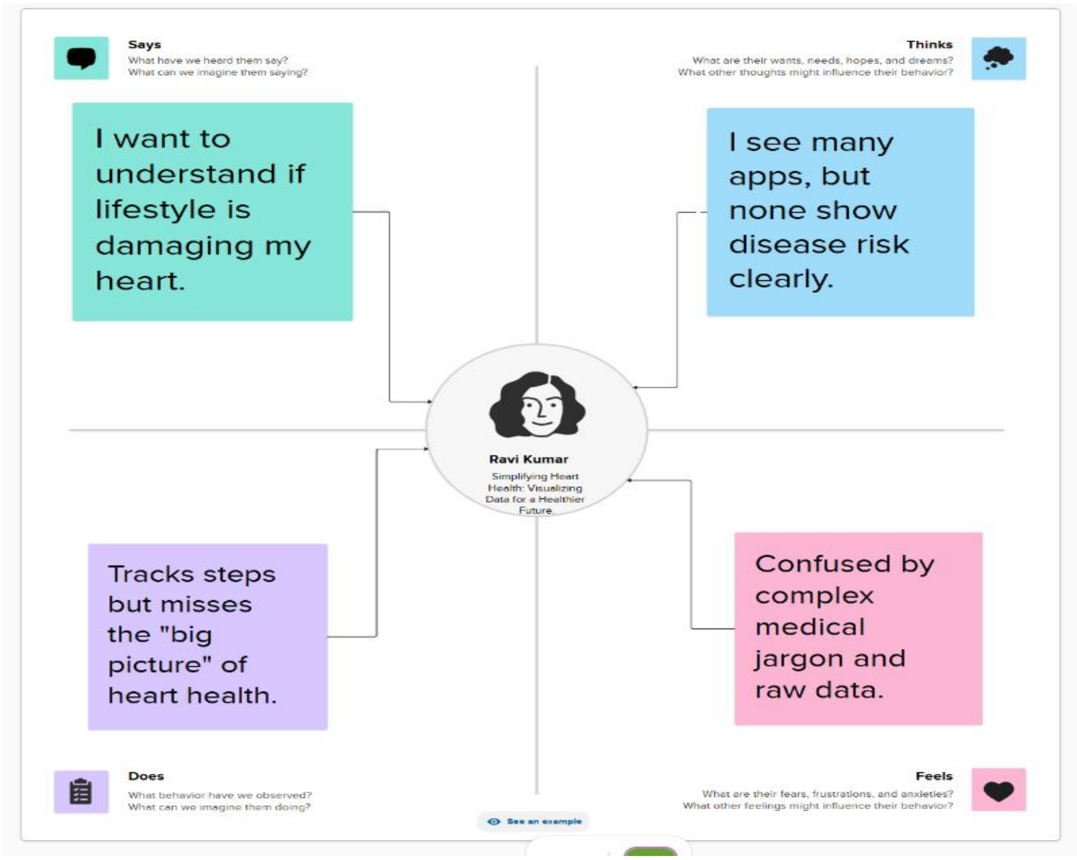
- Clinical Insight: The main goal is to help healthcare professionals understand complex medical data quickly. For example, how factors like BMI, smoking, and diabetes together increase heart disease risk.
- Early Awareness: By creating custom Risk Scores, the project not only shows past data but also provides useful insights that may help in early prevention of heart disease.
- Easy Access: I designed a simple and user-friendly web interface so that even non-technical users can filter and understand clinical data easily.

2. IDEATION PHASE

1.1 Problem Statement
1.2 Empathy map
1.3 Brainstorming

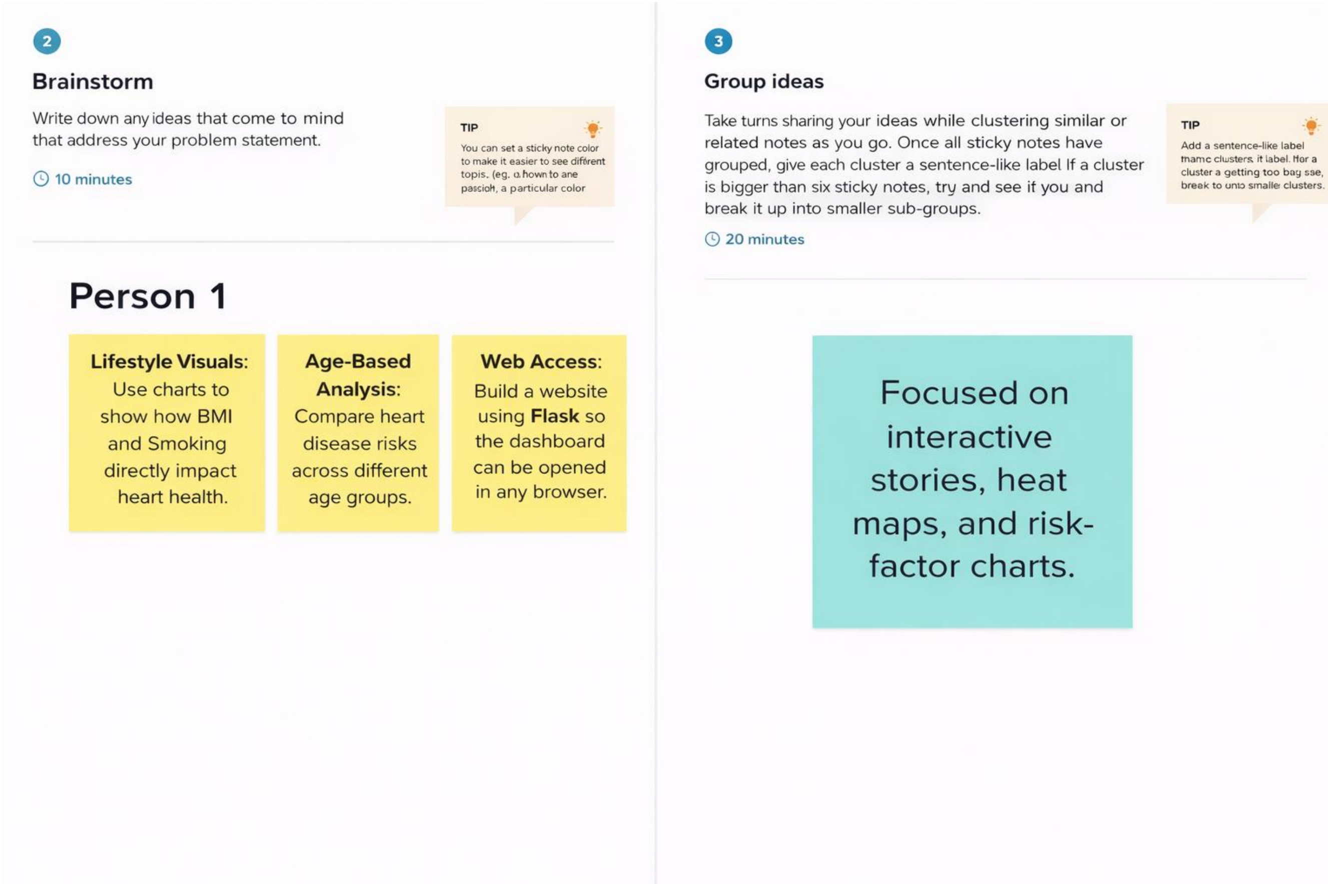
Problem Statement:

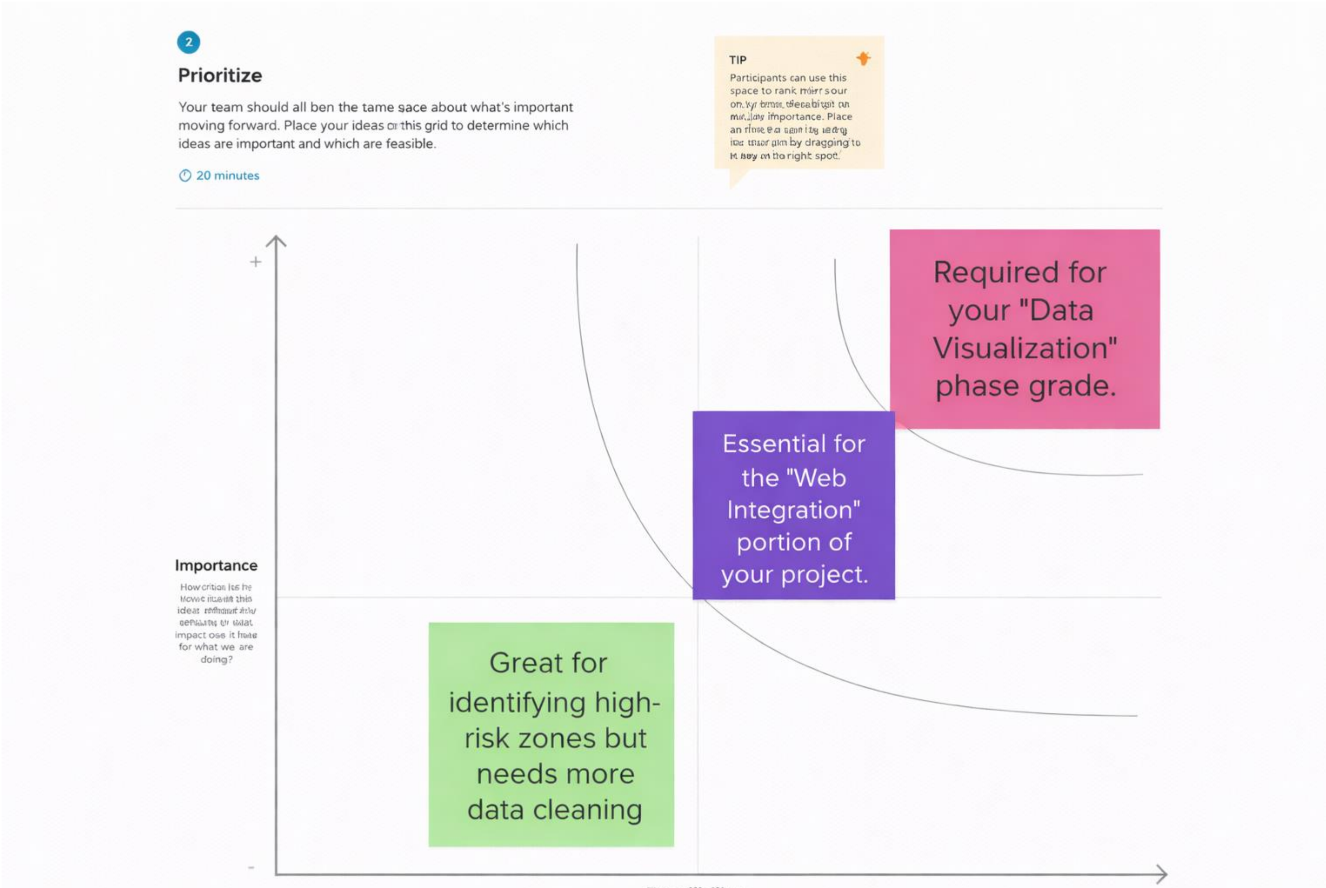
Empathy map:



Says
What have we heard them say?
What can we imagine them saying?
Thinks
What are their wants, needs, hopes, and dreams?
What other thoughts might influence their behavior?
I want to understand if lifestyle is damaging my heart.
I see many apps, but none show disease risk clearly.
Ravi Kumar
Simplifying Heart Health: Visualizing Data for a Healthier Future
Tracks steps but misses the "big picture" of heart health.
Confused by complex medical jargon and raw data.
Does
What behavior have we observed?
What can we imagine them doing?
Feels
What are their fears, frustrations, and anxieties?
What other feelings might influence their behavior?
See an example

Brainstorming:





2
Prioritize
Your team should all ben the tame sace about what's important moving forward. Place your ideas cr this grid to determine which ideas are important and which are feasible.
20 minutes
TIP
Participants can use this
Required for your "Data Visualization" phase grade.
Essential for the "Web Integration" portion of your project.
Importance
+
-
Great for identifying high-risk zones but needs more data cleaning

2. REQUIREMENT ANALYSIS

2.1 Customer Journey map
2.2 Solution Requirement
2.3 Data Flow Diagram
2.4 Technology Stack

Customer Journey Map:

Scenario: [Existing experience through a product or service]	**Entice** How does someone become aware of this service?	**Enter** What do people experience as they begin the process?	**Engage** In the core moments in the process, what happens?	**Exit** What do people typically experience as the process finishes?	**Extend** What happens after the experience is over?
Experience steps What does the person (or people) at the center of this scenario typically experience in each step?	Realizing current medical charts are too hard to read.	Opening the web browser to access your local application.	Exploring the interactive charts and the **Tableau Story**.	Completing the analysis and deciding on a preventive health plan.	Planning to return when new patient data is added.
Interactions What interactions do they have at each step along the way? • **People:** Who do they see or talk to? • **Places:** Where are they? • **Things:** What digital touchpoints or physical objects do they use?	Seeing my project title: **"Heart Disease Analysis".**	Navigating to 127.0.0.1:5000 on a laptop.	Filtering data by Age, BMI, and Smoking status.	Closing the web app or saving a specific chart view.	Bookmarking the local Flask URL for future use.
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Find a better way to visualize clinical risk factors.	A fast, error-free landing page with a clear "Get Started" button.	Understand which lifestyle habits impact heart health most.	Leave the app with clear, actionable medical insights.	Keep monitoring heart health trends over time.
Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	User finds a professional-looking project title: **"CardioPulse".**	The Flask app loads quickly at 127.0.0.1:5000.	**Tableau Story** makes complex risk factors easy to understand.	User successfully identifies their heart health risk level.	User feels empowered to track their health regularly.
Negative moments What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?	Confusion if the project description is too technical.	Users might find it hard to navigate if there are no clear buttons.	Too many charts on one page can feel overwhelming.	Frustration if they cannot save or download the charts.	Data becomes "stale" if the MySQL database isn't updated.
Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Create a video demo showing how easy the dashboard is to use.	Add a "Guided Tour" feature for first-time users.	Use a cleaner layout with fewer charts per row for better focus.	Add a "Download Report" button for patients to take to their doctor.	Automate data uploads so the dashboard shows the latest clinical trends.

See an example

Solution Requirements:

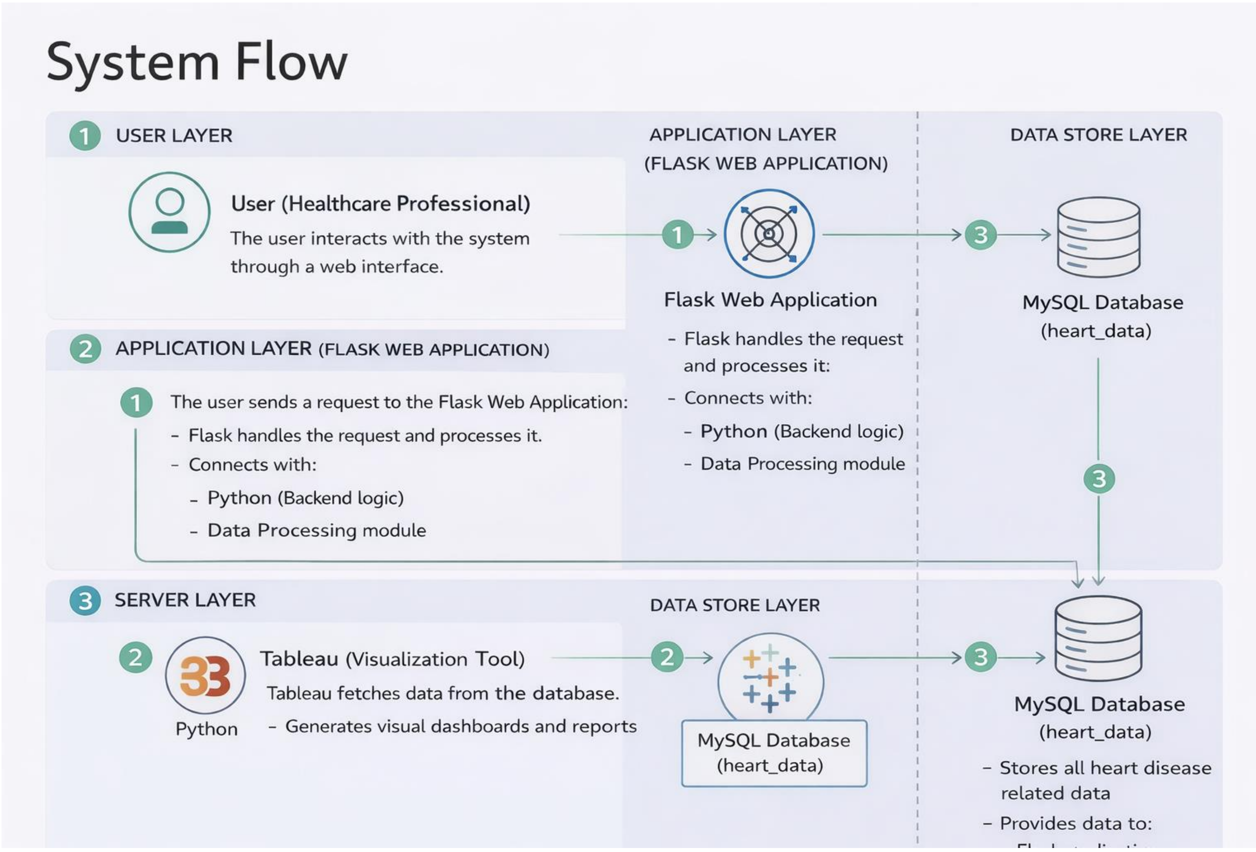
Functional Requirements: Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Extraction	Perform SQL queries on the heart_new2 table to extract clinical data.
FR-2	Data Visualization	Create an interactive Tableau Story with 10+ unique charts showing risk factors.
FR-3	Web Integration	Host the dashboard within a Flask web app using a Bootstrap template.
FR-4	Risk Analysis	Allow users to filter data by Age, BMI, and Smoking status to see specific risk levels.

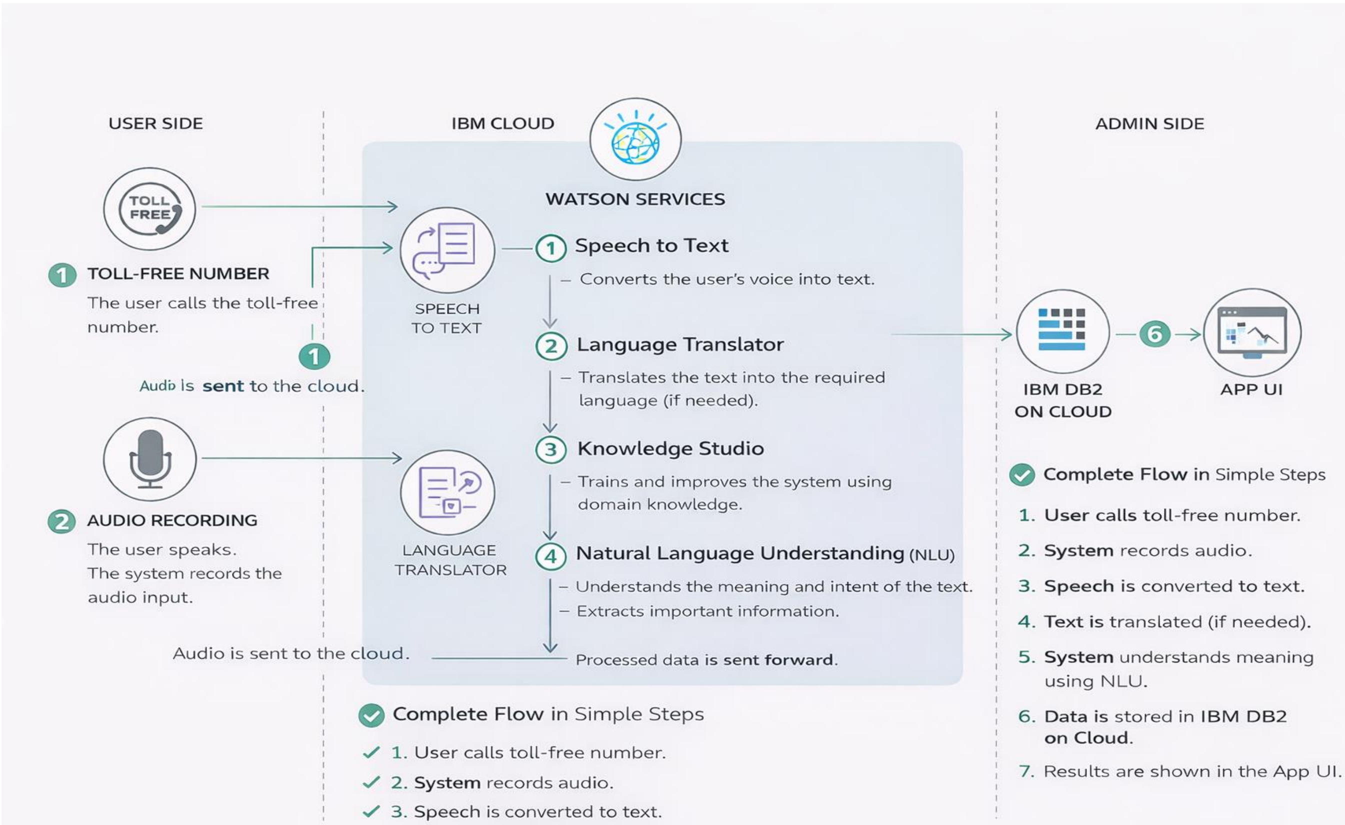
Non-functional Requirements: Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Simple UI for doctors to navigate the dashboard easily.
NFR-2	**Security**	Protect patient data stored in the MySQL database.
NFR-3	**Reliability**	Ensure stable connections between Flask and the database.
NFR-4	**Performance**	Dashboard and charts must load in under 3 seconds.
NFR-5	**Availability**	Accessible 24/7 on the local host server (127.0.0.1:5000).
NFR-6	**Scalability**	Handle larger heart datasets as patient records grow.

Data Flow Diagram:



Technology Stack:



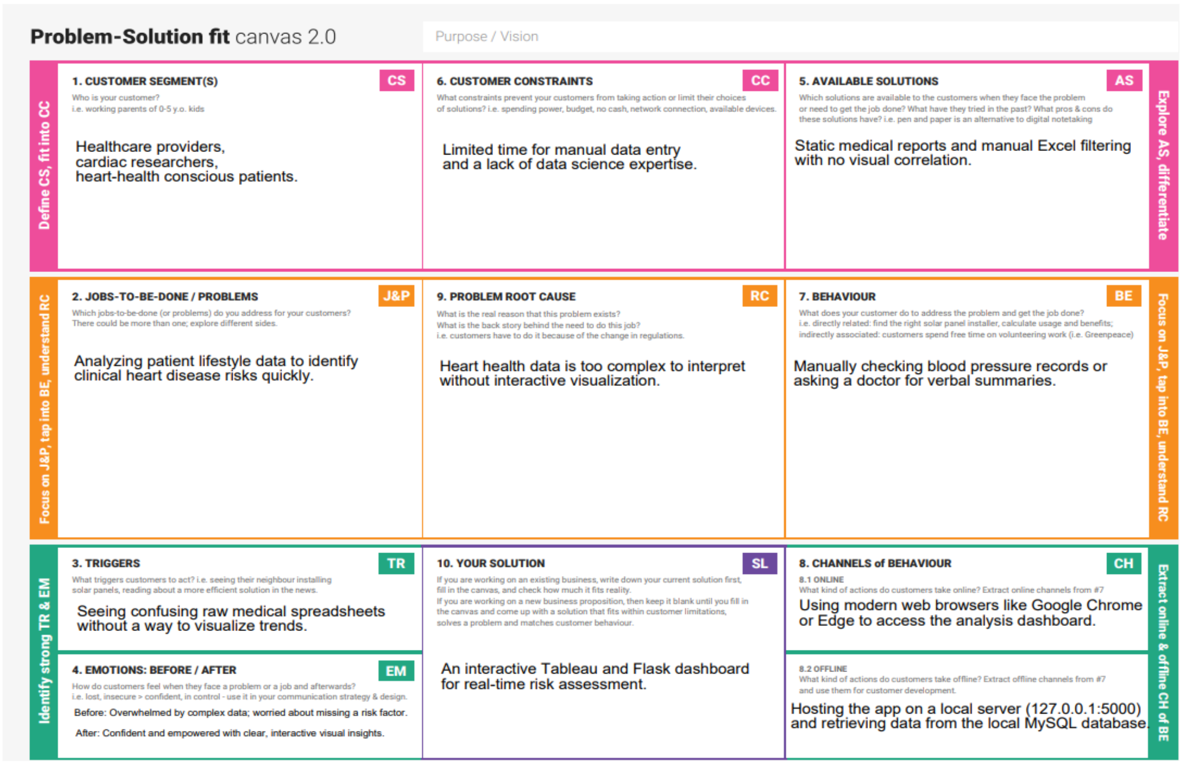
3. PROJECT DESIGN

3.1 Problem Solution Fit

3.2 Proposed Solution

3.3 Solution Architecture

Problem Solution Fit:



Problem-Solution fit canvas 2.0

Purpose / Vision

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

Healthcare providers,
cardiac researchers,
heart-health conscious patients.

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Limited time for manual data entry
and a lack of data science expertise.

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Static medical reports and manual Excel filtering
with no visual correlation.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

Analyzing patient lifestyle data to identify
clinical heart disease risks quickly.

9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

Heart health data is too complex to interpret
without interactive visualization.

7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Manually checking blood pressure records or
asking a doctor for verbal summaries.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

Seeing confusing raw medical spreadsheets
without a way to visualize trends.

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

Before: Overwhelmed by complex data; worried about missing a risk factor.

After: Confident and empowered with clear, interactive visual insights.

10. YOUR SOLUTION SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

An interactive Tableau and Flask dashboard
for real-time risk assessment.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

Using modern web browsers like Google Chrome
or Edge to access the analysis dashboard.

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

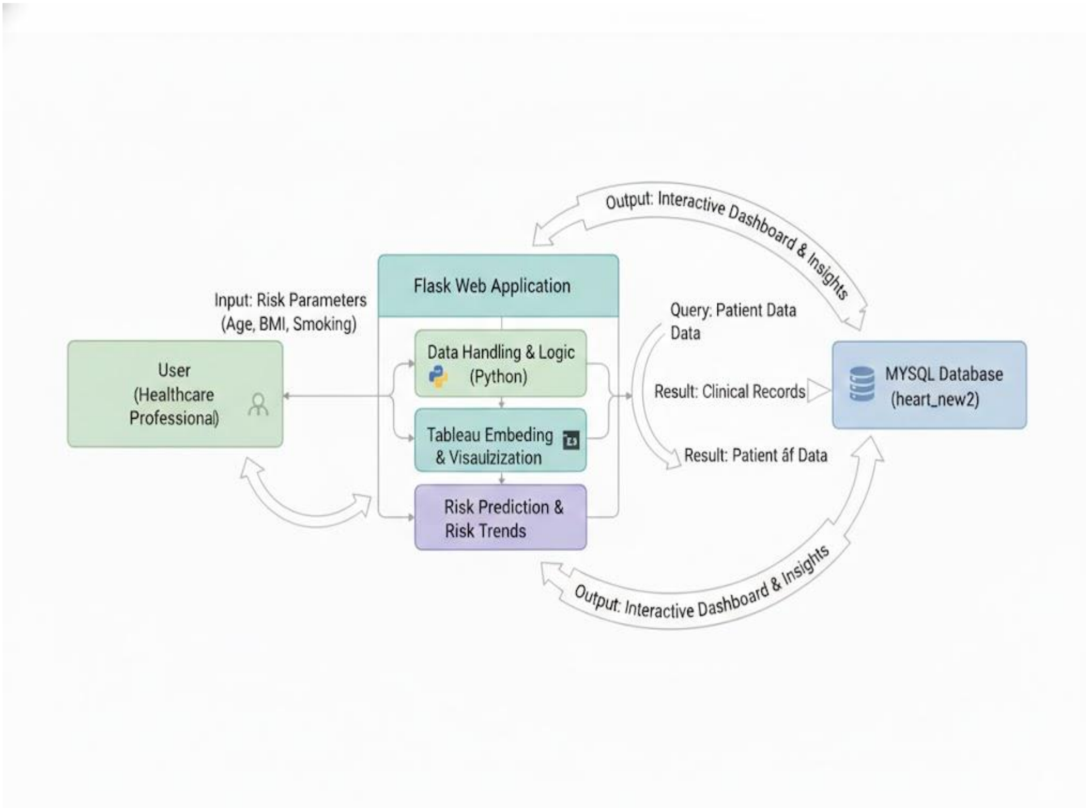
Hosting the app on a local server (127.0.0.1:5000)
and retrieving data from the local MySQL database.

Extract online & offline CH of BE

Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Difficulty in identifying heart disease risks from complex, raw clinical data.
2.	Idea / Solution description	Interactive Flask-web app with 10+ Tableau-driven heart risk visualizations.
3.	Novelty / Uniqueness	Real-time MySQL integration with Tableau Stories for immediate clinical insights.
4.	Social Impact / Customer Satisfaction	Promotes early heart risk detection and builds patient trust through visual data.
5.	Business Model (Revenue Model)	B2B subscription for clinics; free public version for health awareness.
6.	Scalability of the Solution	Modular architecture allows adding kidney, diabetes, or other clinical datasets.

Solution Architecture:



4. PROJECT PLANNING & SCHEDULING

4.1 Project Planning
4.2 Project Milestones & Tasks
4.3 Sprint Delivery Plan
4.4 Project Progress Tracking

Project Planning:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Search for Heart Disease Data.	3	High	Uday
Sprint-1	Database Integration	USN-2	Import heart disease dataset into MySQL and connect to Tableau.	1	High	Uday
Sprint-1	Visualization	USN-3	Build 8+ Heart Disease Visualizations.	3	High	Uday
Sprint-2	Analysis	USN-4	Create the final interactive Dashboard.	5	High	Uday
Sprint-2	Integration	USN-5	Build Tableau Story & Web Integration.	5	High	Uday
Sprint-3	Project Admin	USN-6	Work on Phase templates (Mural/Word).	3	Medium	Uday
Sprint-3	Project Admin	USN-7	Final Documentation & PDF creation.	3	Medium	Uday
Sprint-3	Project Admin	USN-8	Record and edit the Project Demo Video.	2	High	Uday

Project Tracker, Velocity:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	14	5 Days	04 Feb 2026	08 Feb 2026	14	08 Feb 2026
Sprint-2	18	5 Days	09 Feb 2026	13 Feb 2026	18	13 Feb 2026
Sprint-3	08	5 Days	14 Feb 2026	20 Feb 2026	08	20 Feb 2026

Velocity:

$$AV = \textit{sprint duration} / \textit{velocity} = 14 / 5 = 2.8$$

5. FUNCTIONAL AND PERFORMANCE TESTING

5.1 Performance Testing

Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	I successfully connected my clinical dataset to MySQL and Tableau. I verified that all 4,500 records were correctly imported by running row-count queries in MySQL Workbench and checking the data pane in Tableau to ensure every column was visible and ready for analysis.
2.	Data Preprocessing	I performed data cleaning by running SQL DELETE commands to remove invalid records, such as patients with a BMI of 0 or negative sleep hours, which would have ruined the accuracy of my charts. In Tableau, I standardized the data by assigning proper data types (e.g., Number, String) and renaming columns to make them easier for a healthcare professional to understand.
3.	Utilization of Filters	I implemented interactive filters, such as Age Category, Sex, and Smoking Status, to make the dashboard functional for doctors. These filters allow a user to dynamically update all 10 visualizations at once to see risk trends for specific patient groups, like "Males aged 18-24".
4.	Calculation fields Used	I created three specific calculation fields in Tableau to add deeper clinical insights to my project:

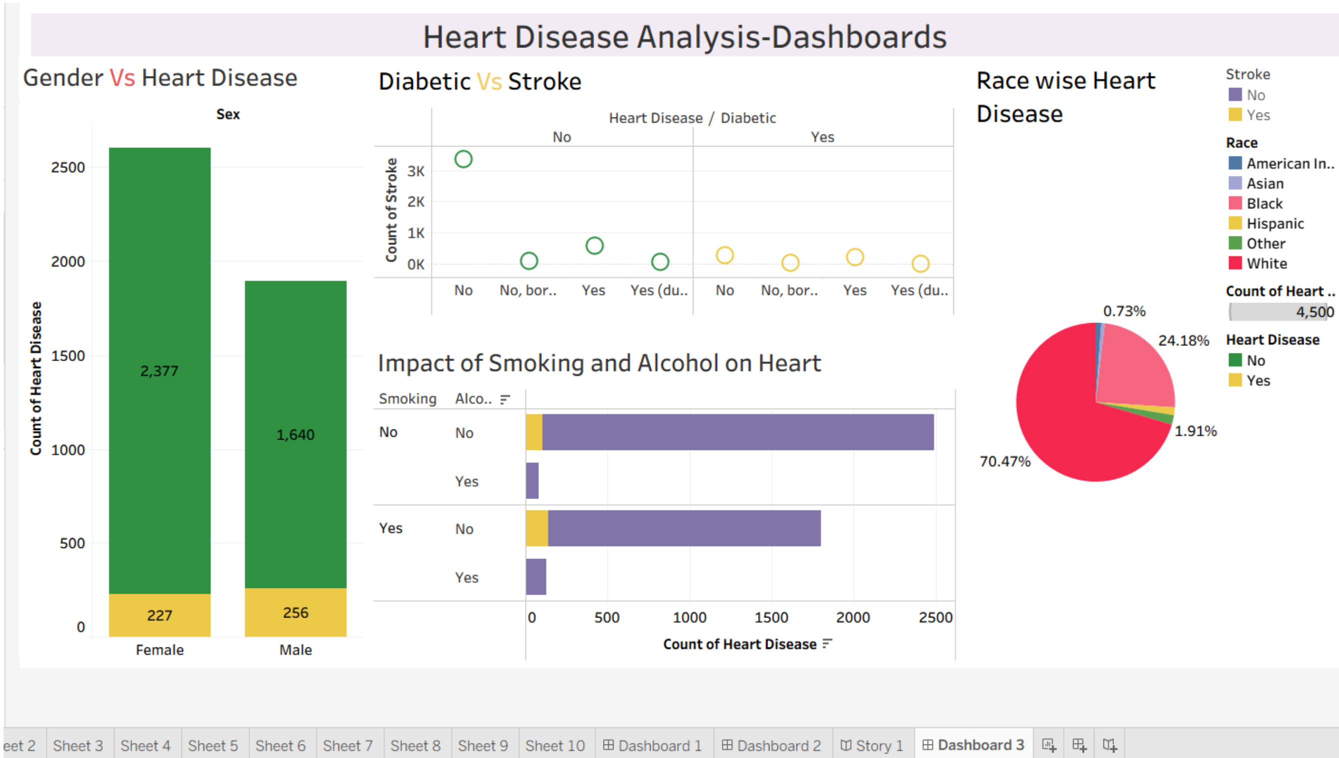
		1) BMI Categories: I transformed raw BMI numbers into easy-to-read groups like "Underweight," "Normal," and "Obese". **Formula:** IF [BMI] < 18.5 THEN "Underweight" ELSEIF [BMI] < 25 THEN "Healthy Weight" ELSEIF [BMI] < 30 THEN "Overweight" ELSE "Obese" END **2) Risk Scores:** I developed a custom score to categorize patients based on their heart disease probability. **Formula**: (IF [Smoking] = 'Yes' THEN 1 ELSE 0 END) + (IF [Stroke] = 'Yes' THEN 2 ELSE 0 END) + (IF [Diabetic] = 'Yes' THEN 1 ELSE 0 END) **3) Percentage Changes:** I created this field to calculate the proportional difference in clinical metrics across different patient groups. **Formula:** (IF [Smoking] = 'Yes' THEN 1 ELSE 0 END) + (IF [Stroke] = 'Yes' THEN 2 ELSE 0 END) + (IF [Diabetic] = 'Yes' THEN 1 ELSE 0 END)

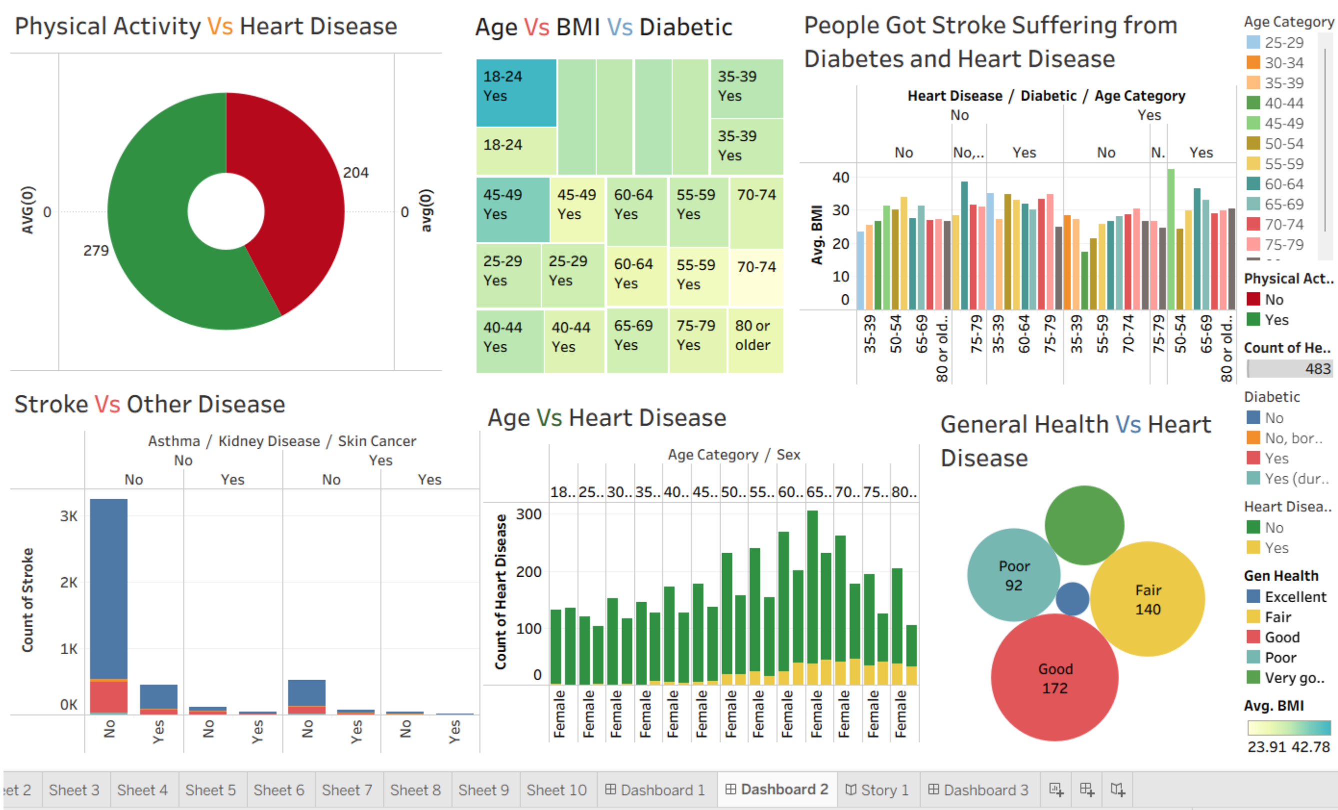
5.	Dashboard design	I built two comprehensive dashboards that integrate 10 distinct visualizations. These dashboards combine various chart types, such as bar charts, pie charts, and heatmaps, to provide a clear overview of clinical heart disease risks. The 10 visualizations included in my analysis are: 1. Gender Vs Heart Disease 2. Age Vs Heart Disease 3. Diabetic Vs Stroke 4. Impact of Smoking and Alcohol on Heart 5. Stroke Vs Other Disease 6. Race wise heart disease 7. General Health Vs Heart Disease 8. Physical Activity Vs Heart Disease 9. Age Vs BMI Vs Diabetic 10. People Got Stroke Suffering from Diabetes and Heart Disease
6	Story Design	I organized my analysis into a narrative flow using **10 individual visualizations**. This allows a viewer to see the project's progress from basic demographic data to complex clinical health risks. The sequence of my analysis is as follows: 1. **Gender Vs Heart Disease**: Identifying risk based on gender. 2. **Age Vs Heart Disease**: Analysing how age affects heart health. 3. **Diabetic Vs Stroke**: Examining the link between diabetes and stroke. 4. **Impact of Smoking and Alcohol on Heart**: Evaluating lifestyle choices.

		5. **Stroke Vs Other Disease**: Comparing stroke rates with other health conditions. 6. **Race wise heart disease**: Studying heart disease trends across different races. 7. **General Health Vs Heart Disease**: Seeing how personal health ratings correlate with disease. 8. **Physical Activity Vs Heart Disease**: Measuring the impact of exercise. 9. **Age Vs BMI Vs Diabetic**: A three-way analysis of age, body mass, and diabetes. 10. **People Got Stroke Suffering from Diabetes and Heart Disease**: A final look at high-risk patients with multiple conditions.

6. RESULTS

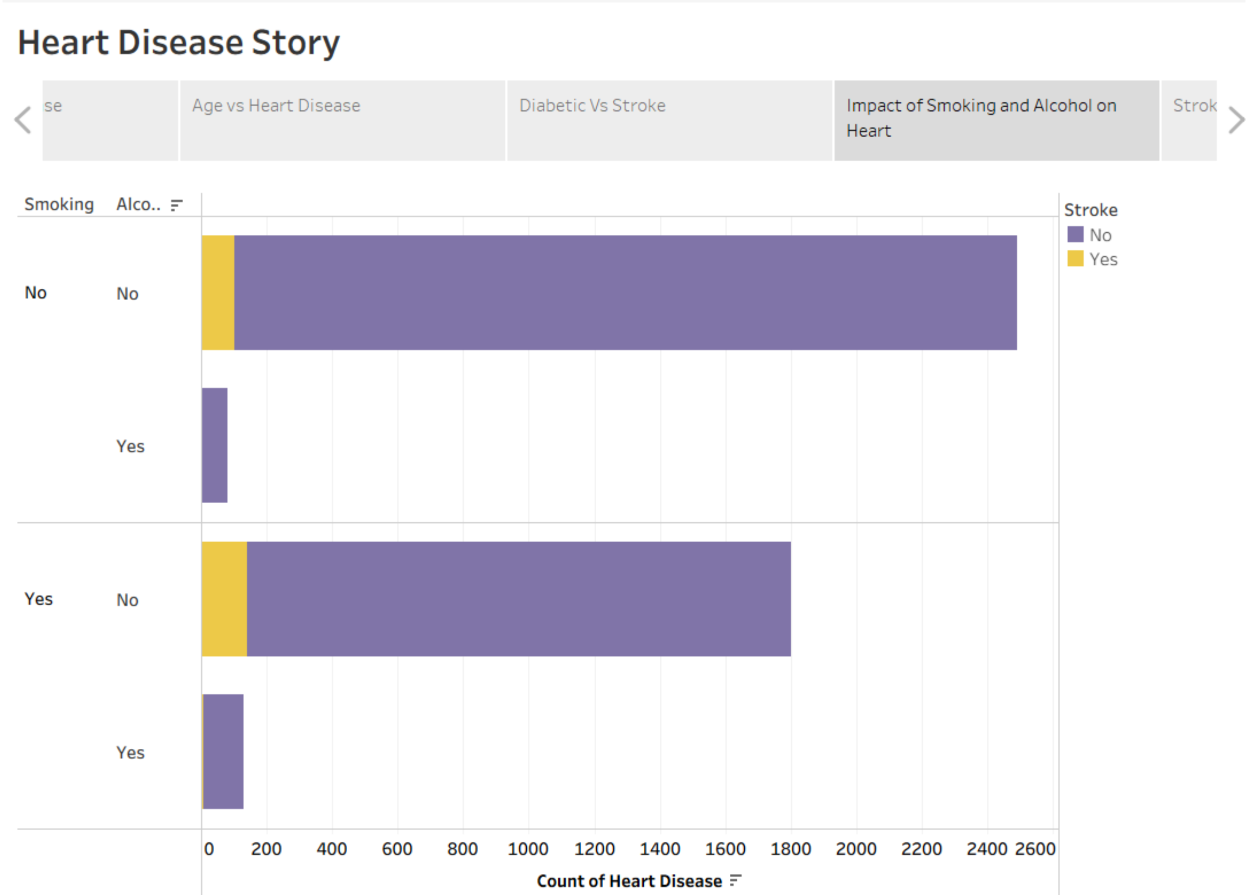
Dashboard Screenshots:



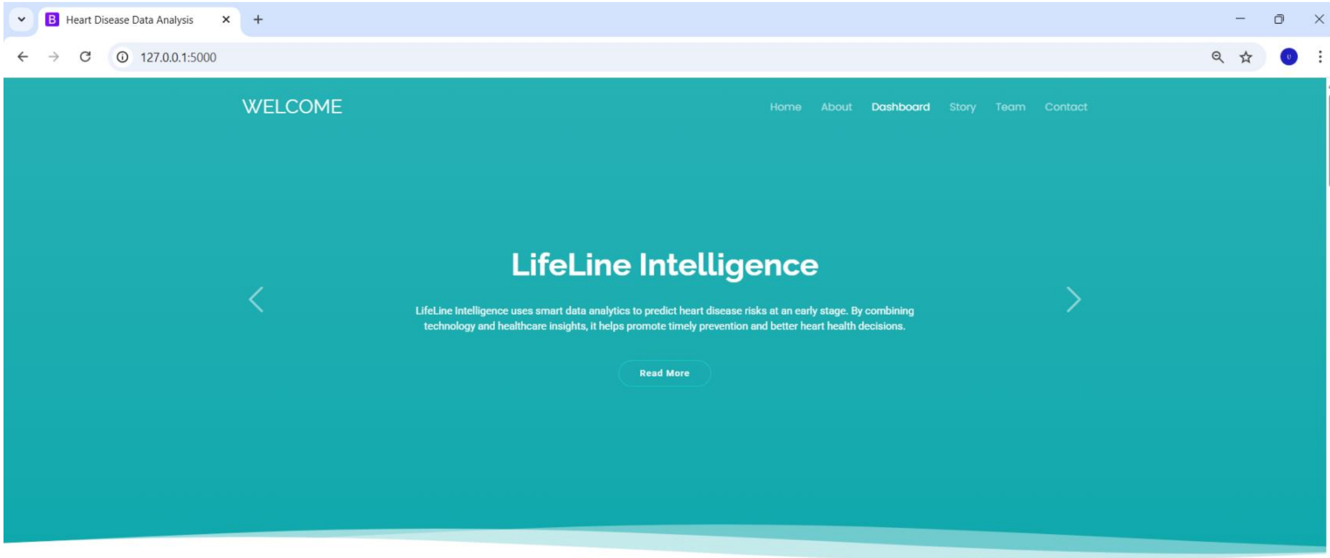


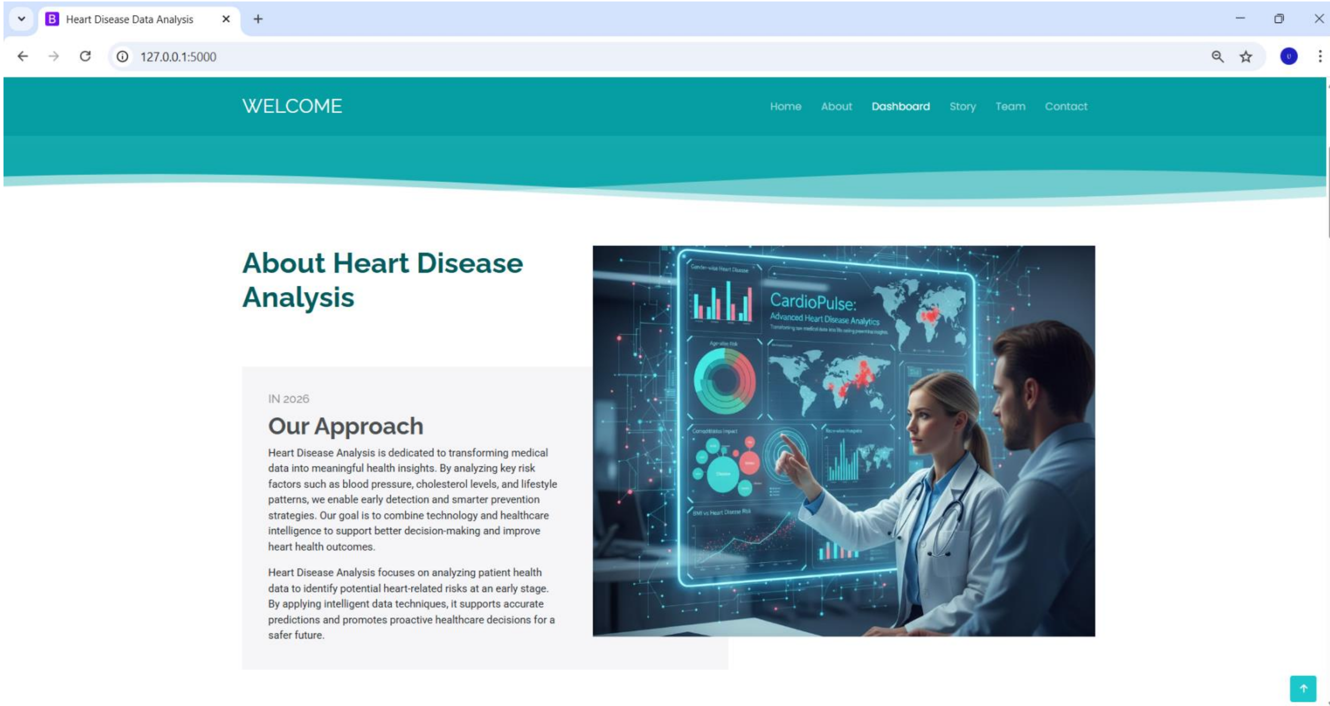
Physical Activity Vs Heart Disease
AVG(0)
avg(0)
0
0
204
279
Age Vs BMI Vs Diabetic
18-24 Yes
18-24
35-39 Yes
35-39 Yes
45-49 Yes
45-49 Yes
60-64 Yes
55-59 Yes
70-74
25-29 Yes
25-29 Yes
60-64 Yes
55-59 Yes
70-74
40-44 Yes
40-44 Yes
65-69 Yes
75-79 Yes
80 or older
People Got Stroke Suffering from Diabetes and Heart Disease
Heart Disease / Diabetic / Age Category
Avg. BMI
Age Category
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
Physical Act..
No
Yes
Count of He..
483
Stroke Vs Other Disease
Asthma / Kidney Disease / Skin Cancer
Count of Stroke
Age Vs Heart Disease
Age Category / Sex
Count of Heart Disease
General Health Vs Heart Disease
Poor 92
Fair 140
Good 172
Diabetic
No
No, bor..
Yes
Yes (dur..
Heart Disea..
No
Yes
Gen Health
Excellent
Fair
Good
Poor
Very go..
Avg. BMI
23.91 42.78
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Sheet 9
Sheet 10
Dashboard 1
Dashboard 2
Story 1
Dashboard 3

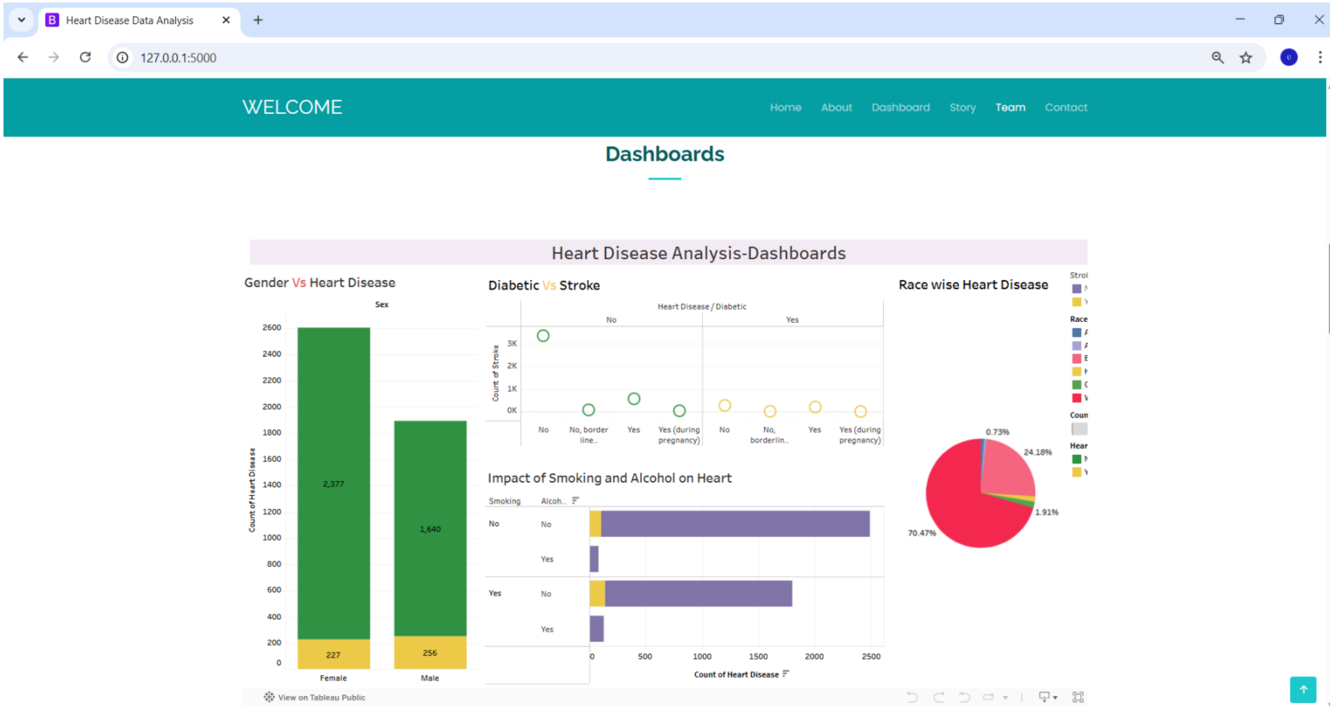
Story Screenshot:



Web Integration Screenshots:




Heart Disease Data Analysis
127.0.0.1:5000
WELCOME
Home
About
Dashboard
Story
Team
Contact
About Heart Disease Analysis
IN 2026
Our Approach
Heart Disease Analysis is dedicated to transforming medical data into meaningful health insights. By analyzing key risk factors such as blood pressure, cholesterol levels, and lifestyle patterns, we enable early detection and smarter prevention strategies. Our goal is to combine technology and healthcare intelligence to support better decision-making and improve heart health outcomes.
Heart Disease Analysis focuses on analyzing patient health data to identify potential heart-related risks at an early stage. By applying intelligent data techniques, it supports accurate predictions and promotes proactive healthcare decisions for a safer future.
CardioPulse:
Advanced Heart Disease Analytics


Heart Disease Data Analysis
127.0.0.1:5000
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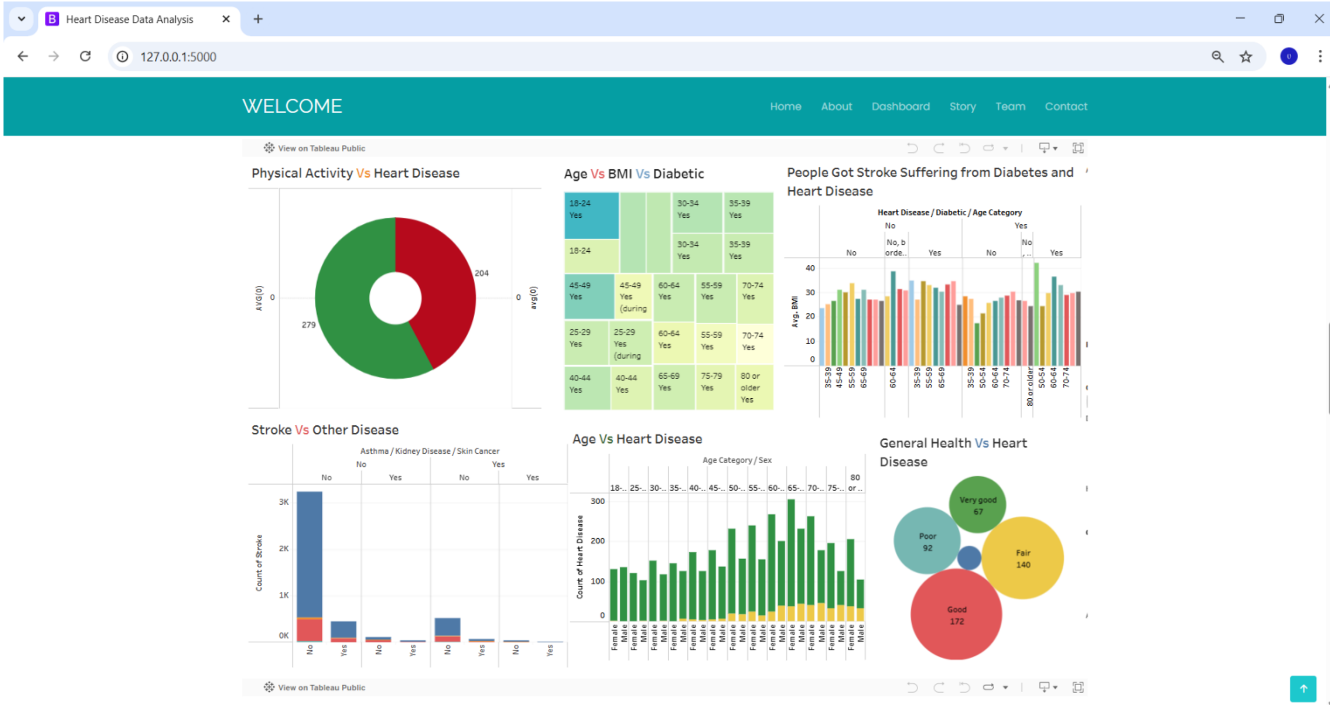
Dashboards
Heart Disease Analysis-Dashboards
Gender Vs Heart Disease
Diabetic Vs Stroke
Race wise Heart Disease
Impact of Smoking and Alcohol on Heart
Count of Heart Disease
View on Tableau Public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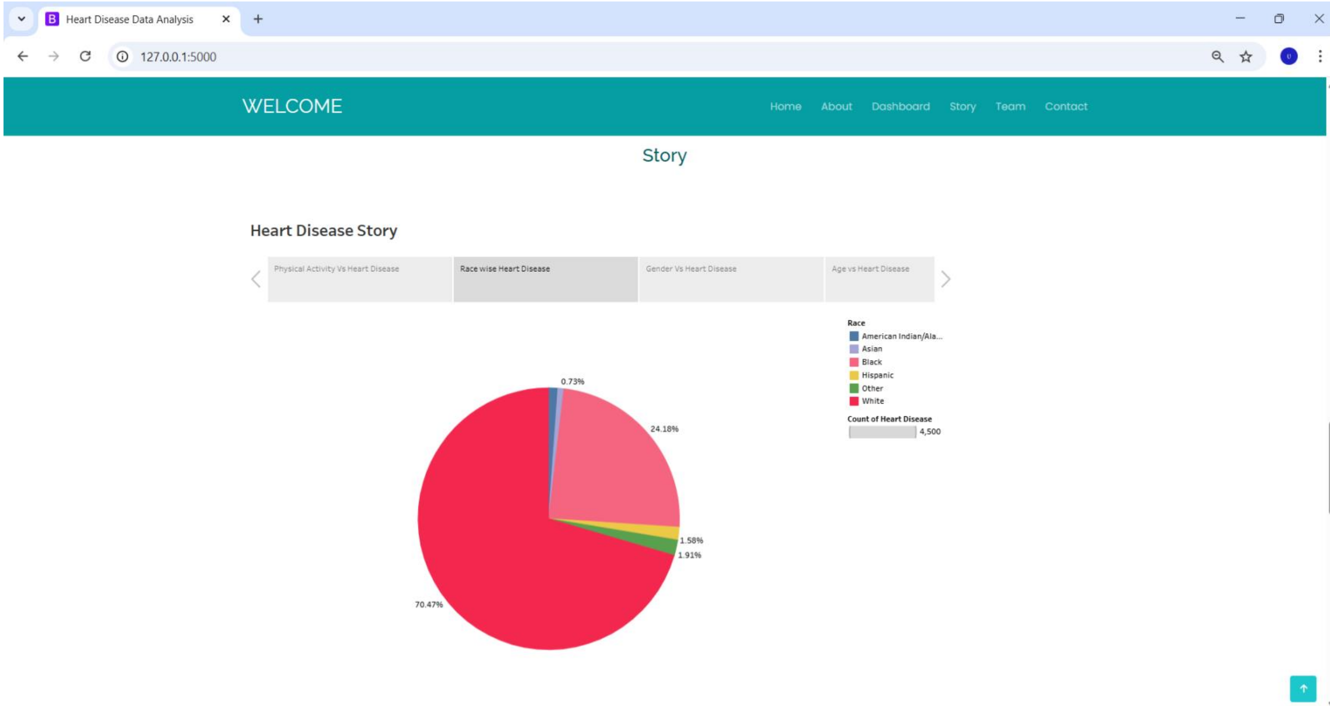
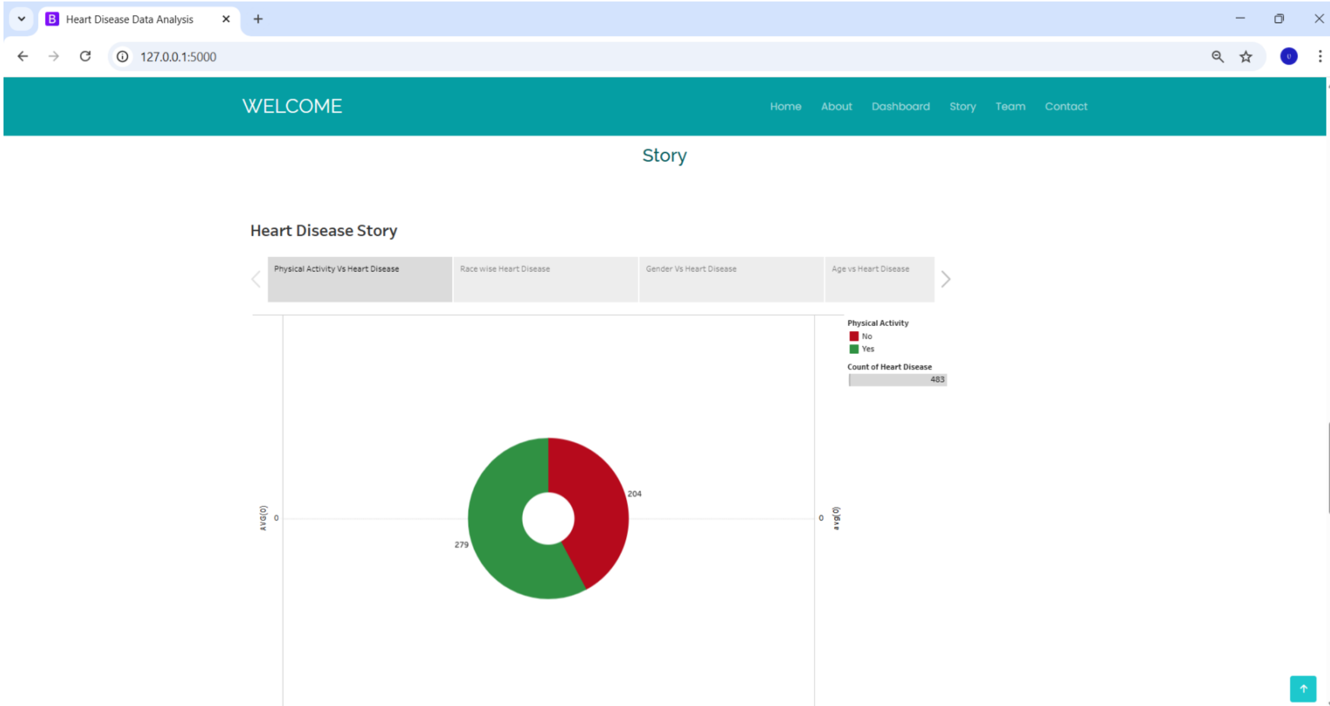
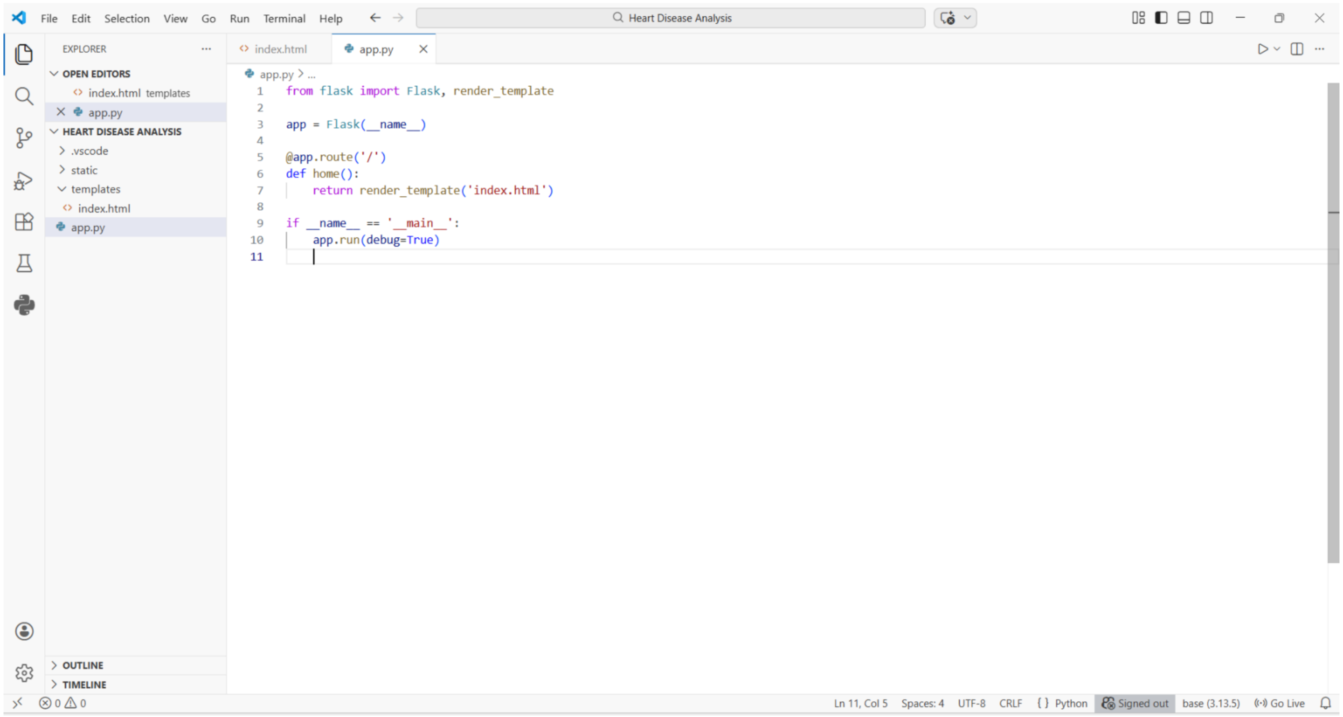
View on Tableau Public
Physical Activity Vs Heart Disease
Age Vs BMI Vs Diabetic
People Got Stroke Suffering from Diabetes and Heart Disease
Heart Disease / Diabetic / Age Category
Avg. BMI
Stroke Vs Other Disease
Asthma / Kidney Disease / Skin Cancer
Count of Stroke
Age Vs Heart Disease
Age Category / Sex
Count of Heart Disease
General Health Vs Heart Disease
Very good
67
Poor
92
Fair
140
Good
172
View on Tableau Public


Heart Disease Data Analysis
127.0.0.1:5000
WELCOME
Home
About
Dashboard
Story
Team
Contact
Story
Heart Disease Story
Physical Activity Vs Heart Disease
Race wise Heart Disease
Gender Vs Heart Disease
Age vs Heart Disease
0.73%
24.18%
1.58%
1.91%
70.47%
Race
American Indian/Ala...
Asian
Black
Hispanic
Other
White
Count of Heart Disease
4,500

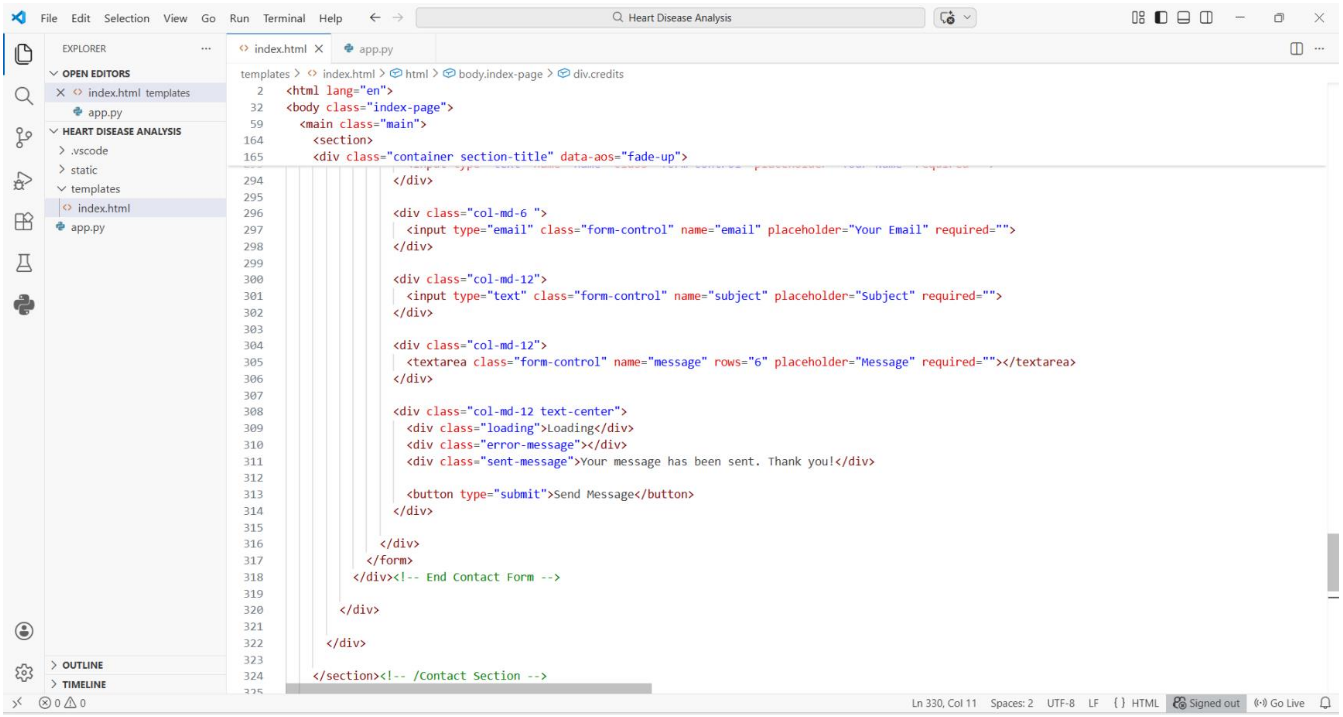

Heart Disease Data Analysis
127.0.0.1:5000
WELCOME
Home
About
Dashboard
Story
Team
Contact
Story
Heart Disease Story
Physical Activity Vs Heart Disease
Race wise Heart Disease
Gender Vs Heart Disease
Age vs Heart Disease
Physical Activity
No
Yes
Count of Heart Disease
483
204
279

VS Code Screenshots:

Python Flask code:

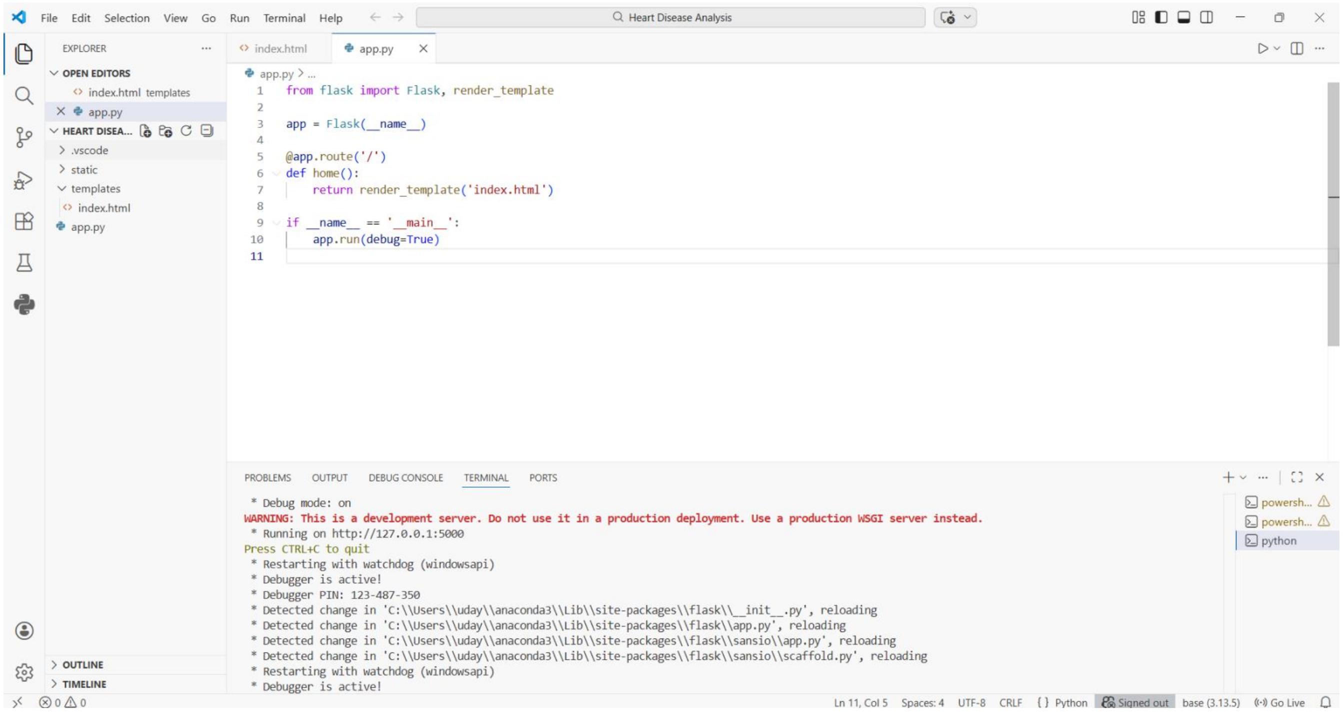


Index.html Code:

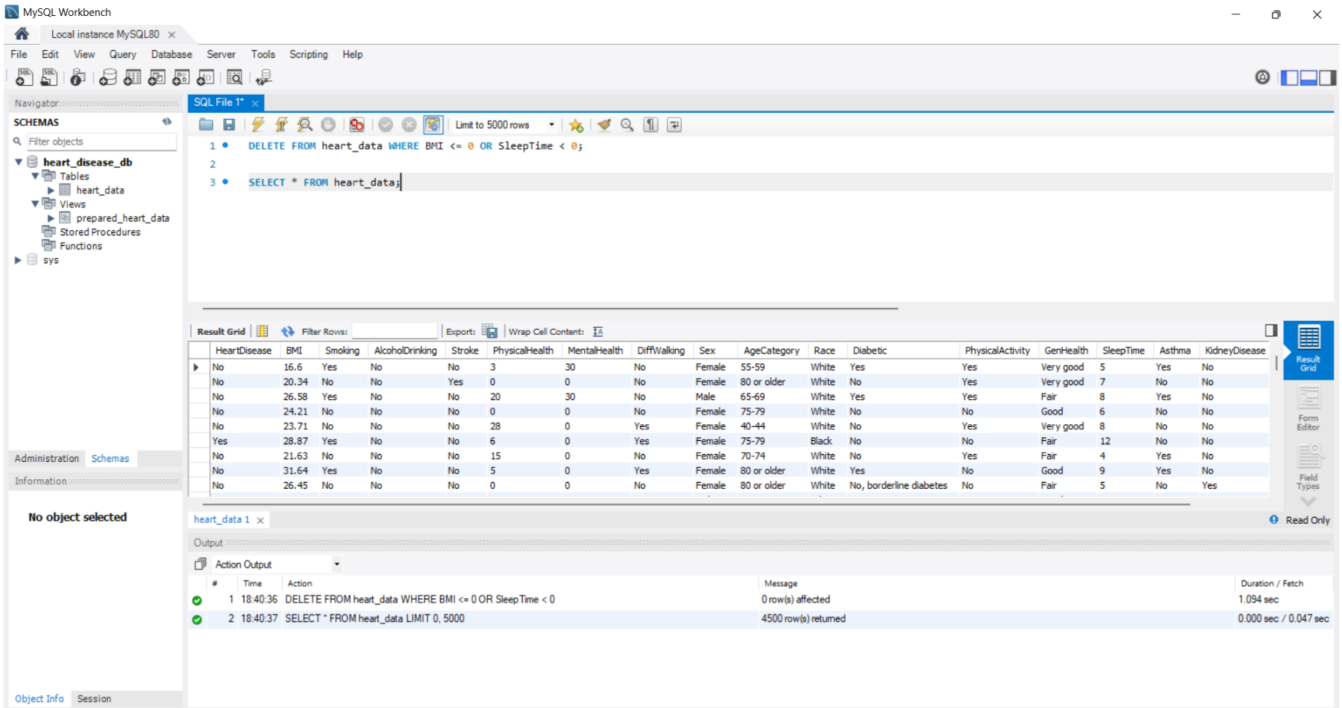


```html
<html lang="en">
<body class="index-page">
  <main class="main">
    <section>
    <div class="container section-title" data-aos="fade-up">
              </div>

              <div class="col-md-6 ">
                <input type="email" class="form-control" name="email" placeholder="Your Email" required="">
              </div>

              <div class="col-md-12">
                <input type="text" class="form-control" name="subject" placeholder="Subject" required="">
              </div>

              <div class="col-md-12">
                <textarea class="form-control" name="message" rows="6" placeholder="Message" required=""></textarea>
              </div>

              <div class="col-md-12 text-center">
                <div class="loading">Loading</div>
                <div class="error-message"></div>
                <div class="sent-message">Your message has been sent. Thank you!</div>

                <button type="submit">Send Message</button>
              </div>

            </div>
          </form>
        </div><!-- End Contact Form -->

      </div>

    </div>

  </section><!-- /Contact Section -->
```

Python Flask Output:



MySQL Operations Output:



8. ADVANTAGES & DISADVANTAGES

Advantages

- Large Data Size: My project uses 4,500 records. Because of this large dataset, the health trends and results are more reliable.
- Easy to Use: I added filters like Age and Sex, so anyone can search and understand the data easily without technical knowledge.
- Better Medical Insights: I created new fields like Risk Scores and BMI Categories. These give doctors more meaningful information than the original raw data.
- Clear Visual Explanation: I created 10 charts and an 8-step story to explain the data clearly. This makes complex medical information simple and easy to understand.

Disadvantages

- Based on Old Data: The project uses past records, so it cannot predict heart disease for new patients in real-time.
- Needs Internet: Since I used Tableau Public and a web-based Flask app, an internet connection is required to view the dashboards.
- Manual Updates Required: When new patient data is added, I need to manually clean and update the MySQL database again.
- Limited Factors: The analysis only includes the data available in the dataset. It does not consider other important factors like genetics, diet, or lifestyle habits.

9. CONCLUSION

The Heart Disease Analysis project shows how data science can turn raw medical records into useful health insights. I connected a MySQL database with Tableau visualizations and a Flask web application to create an interactive system.

After cleaning and preparing 4,500 records, I built 10 visualizations and 2 dashboards to provide accurate and meaningful analysis.

By using filters like Age and Sex, the platform becomes easy for healthcare professionals to use. This project proves that well-organized data and clear visual storytelling can improve health awareness and help in early detection of heart disease.

10. FUTURE SCOPE

In the future, this project can be improved by adding Machine Learning models to predict heart disease risk for new patients in real-time.

More data sources like genetic history and wearable device data can also be included to give a complete health overview.

The web application can be converted into a mobile app for easier access by doctors. An automatic data system can also be added so that the dashboards update instantly when new data is added.

Finally, an AI chatbot can be included to help users understand the medical information in simple language.

11. APPENDIX

- **Tableau Public Link:** https://public.tableau.com/app/profile/puthalapattu.uday.kiran/vizzes
- **Dataset Link:** https://drive.google.com/file/d/1IL6GcTkLYQVFfnR-4LnIzq1B_d8thIWk/view?usp=drivesdk
- **GitHub Link:** https://github.com/udaykiranreddy458-sys/Heart-Disease-Analysis
- **Project Demo:** https://drive.google.com/file/d/1-zGEP81MebKdr6DVvKqYG4Pdxu-L_pUA/view?usp=drivesdk